

Felipe Gutierrez

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[Link to Portfolio](#)

CAREER SUMMARY

- Third-year Mechanical Engineering (Honours) student at UTS with hands-on experience in mechanical design, structural analysis, and SolidWorks CAD.
- Focused on mechanical design and physical systems engineering, with a drive to apply this knowledge to real engineering problems.

TECHNICAL SKILLS

- **SolidWorks (Proficient):** Designed and simulated mechanical systems across multiple subjects, including CFD analysis with SolidWorks Flow Simulation.
- **MATLAB, Python, C++ (Basic):** Applied control and data analysis scripts in Introduction to Mechatronics.
- **Mechatronics Integration (Basic):** Combined mechanical, electrical, and software components in a lab setting to enhance existing systems.
- **Microsoft Office (Proficient):** Skilled in Word, Excel, and PowerPoint for data analysis, documentation, and presentations.
- **Languages:** English (Bilingual proficiency) and Spanish (Bilingual proficiency).

EDUCATION

Bachelor of Mechanical Engineering (Honours)

University of Technology Sydney

Aug 2024 – Nov 2028

EXPERIENCE

Warrane College – Residential Tutor

Sydney, Australia

Feb 2026 – Present

- Provided pastoral care and mentorship to residents, supporting their transition into university life.
- Promoted student wellbeing through regular check-ins, informal counselling, and early intervention.
- Fostered an inclusive residential environment across diverse cultural and academic backgrounds.
- Planned and delivered weekly floor events to encourage engagement and community culture.
- Offered peer guidance and conflict support, reinforcing respectful and responsible college living.

Cake Shop and Coffee, Pasteleria Alvarez Carmona

Coquimbo, Chile

Nov 2018 – Apr 2022

- Worked as a waiter in a customer service-oriented cafe, delivering barista and table service.
- Assisted with inventory management, restocking and organising products to meet demand.

PROJECTS

2026 Weir Warman Design & Build Competition

41059 Mechanical Design Fundamentals Studio 1 | University of Technology Sydney

Autumn 2026 – Present

- Worked in a five-person team to design and build an autonomous robot for the 39th Weir Warman Competition.
- Led the chassis sub-system, from structural design through to final build.
- Served as Research Lead and Quality & Safety Lead, producing risk assessments and SWMS.

Counter-Flow Heat Exchanger Design and Analysis

Autumn 2025

41057 Thermofluids A | University of Technology Sydney

- Worked in a team to design and build a counter-flow heat exchanger, using CFD simulations and the NTU method to analyse thermal performance.

Vertical Axis Wind Turbine – Design, Fabrication and Testing

Spring 2025

41053 Materials and Manufacturing Engineering A | University of Technology Sydney

- Designed, fabricated, and tested a six-blade VAWT in a four-person team, generating measurable electrical output at low wind speeds.
- Designed 3D-printed PETG connectors, fabricated components using UTS workshop equipment, and tested performance on a roller-driven rig.

VOLUNTEERING

Warrane College Volunteering – Little Sisters Nursing Home

Feb 2024 – Present

Randwick, NSW

- Assisted with weekly gardening and maintenance tasks to maintain a welcoming environment.
- Collaborated with volunteers to organise supplies and set up communal areas as needed.
- Built interpersonal and empathy skills by regularly interacting with and offering companionship to elderly residents.
- Committed to a consistent Saturday schedule, demonstrating reliability and community spirit.

EXTRA-CURRICULAR EXPERIENCE

AusIMM Student Member

2026 – Present

Sydney, Australia

- Student member of the Australasian Institute of Mining and Metallurgy, staying connected to the mining and resources industry through resources, events, and professional networks.

President, Chess Club at Warrane College

Jan 2024 – Dec 2024

Sydney, Australia

- Organised weekly sessions and inter-college tournaments, fostering a supportive environment for all skill levels.
- Demonstrated leadership and event management skills while coordinating club activities and promoting engagement.

UNSW Intercollege Cup Competitor, Warrane College

Apr 2023 – Present

Sydney, Australia

- Represented Warrane College in OzTag, football, and basketball, demonstrating teamwork, resilience, and sportsmanship.
- Balanced athletic commitments with academic responsibilities, demonstrating time management and dedication.

REFEREES

Available upon request.